Payment provider API readiness

Status: STAGING CONTRACT - DO NOT USE AS A PRODUCTION ACTIVATION NOTICE

The target production origin remains:

https://billing.superoptic.net

The pre-production acceptance origin is:

https://mgt.superoptic.net

The payment API routes described in this directory become production routes only after SuperOptic sends a written activation notice. A DNS record, login page, or successful TLS handshake alone is not an activation notice.

Credential model

Each provider receives its own:

- public key ID;
- one-time private secret used in the bearer credential and HMAC-SHA256 signature;
- production source-IP allowlist;
- test and production activation dates.

Emanat and United Payment credentials are not interchangeable. No credential is stored in this repository or embedded in the documentation.

The public key ID is an identifier, not an RSA/ECDSA public key. The private secret is shown once and only its Argon2id hash is retained by SuperOptic. If a provider requires asymmetric RSA/ECDSA exchange, that is a different authentication profile and must be agreed before activation.

Required provider information

Before production activation, each provider must return:

- legal provider name and technical contact;
- test and production source IP addresses;
- maximum timeout and retry policy;
- confirmation that custom HTTP headers are supported;
- confirmation that HMAC-SHA256 is supported;
- confirmation that the same transaction ID is reused on retries;
- planned test window.

Operational contract

- TLS 1.2 or newer is required.
- Requests time out after 15 seconds on the provider side.
- Payment retries must reuse the same transaction ID and idempotency key.
- A successful payment response means the payment was durably recorded.
- RADIUS/service activation is asynchronous and is exposed through

payment_state.

- Every response returns X-Correlation-ID and a JSON correlation_id.

- Provider credentials must be exchanged through an approved secure channel,

never through email body, chat, source code, or a ticket comment.

SuperOptic Emanat payment API v1

Status: staging contract

Base URL after written activation:

https://billing.superoptic.net/api/provider

Acceptance-test base URL before activation:

https://mgt.superoptic.net/api/provider

Authentication

All requests include:

```
Authorization: Bearer <PUBLIC_KEY_ID>.<PRIVATE_SECRET>
X-Provider-Name: emanat
X-Superoptic-Timestamp: <Unix time in seconds>
X-Superoptic-Signature: <lowercase hex HMAC-SHA256>
X-Correlation-ID: <provider request correlation id>
```

The signature input is the exact byte sequence:

```
<timestamp>\n<raw HTTP request body>
```

PUBLIC_KEY_ID is safe to use as a credential identifier. PRIVATE_SECRET is shown only once when the key is created and must be stored in Emanat's secret manager. The HMAC signing key is the same PRIVATE_SECRET. The timestamp must be within 300 seconds of SuperOptic server time. For GET requests the raw body is empty.

Payment requests also include:

```
Idempotency-Key: emanat:<transaction_id>
Content-Type: application/json
```

1. Customer lookup

```
GET /api/provider/customer-lookup.php?provider=emanat&customer_code=2320025
```

Success:

```
{
  "success": true,
  "provider": "emanat",
  "customer": {
    "customer_code": "2320025",
    "contract_no": "2320025",
    "full_name": "Example Customer",
    "status": "Aktiv",
    "is_active": true,
    "package": "100 Mbps",
    "balance": "0 AZN",
    "balance_amount": 0,
    "remaining_days": 25,
    "expires_at": "2026-08-22 12:00:00"
  },
  "correlation_id": "req-..."
}
```

Unknown customer returns HTTP 404 with code=customer_not_found.

2. Record payment

```
POST /api/provider/payment.php
```

```
{
  "provider": "emanat",
  "terminal_id": "EMN-001",
  "customer_code": "2320025",
  "transaction_id": "EMN-20260728-000001",
  "amount": "25.00",
  "paid_at": "2026-07-28T16:30:00+04:00"
}
```

Success:

```
{
  "success": true,
  "duplicate": false,
  "external_id": "emanat:EMN-20260728-000001",
  "transaction_id": "EMN-20260728-000001",
  "amount": "25.00",
  "payment_state": "service_action_pending",
  "customer": {},
  "correlation_id": "req-..."
}
```

Retrying the exact request with the same transaction ID is idempotent and does not create a second payment. Reusing the key with a different amount or customer is rejected.

3. Payment status

```
GET /api/provider/payment-status.php?provider=emanat&transaction_id=EMN-20260728-000001
```

The payment_state value is one of:

- payment_recorded
- service_action_pending
- service_action_processing
- service_action_completed
- service_action_retrying
- manual_reconciliation_required
- reversed

Unknown transactions return HTTP 404 with code=payment_not_found.

HTTP error summary

HTTP	Code	Meaning
400	unsupported_provider	Provider name is invalid
400	customer_code_required	Customer code is missing
400	transaction_id_required	Transaction ID is missing
400	amount_required	Amount is missing or invalid
401	unauthorized	API credential is invalid
401	credential_revoked	API credential was revoked

HTTP	Code	Meaning
401	credential_expired	API credential expired
401	invalid_timestamp	Timestamp is outside the accepted window
401	invalid_signature	HMAC signature is invalid
403	source_ip_not_allowed	Source IP is outside the allowlist
403	scope_denied	API credential does not have the required scope
404	customer_not_found	Customer does not exist
500	payment_apply_failed	Payment was not committed; retry safely
503	provider_credentials_not_configured	Provider is not activated

Payment provider production cutover checklist

SuperOptic

[ ] billing.superoptic.net routes /api/provider/ to the unified portal.

[ ] /api/health.php returns HTTP 200 through the public TLS origin.

[ ] Emanat and United Payment have separate public key IDs and private secrets.

[ ] Shared legacy provider credentials are disabled.

[ ] Provider production source IPs are configured.

[ ] TLS certificate has at least 14 days remaining.

[ ] Database backup and rollback point are verified.

[ ] Payment outbox worker and timer are active.

[ ] RADIUS action worker is active or deliberately isolated for the test.

[ ] Duplicate payment, payload mismatch, retry, and status tests pass.

[ ] Failed RADIUS action reaches retry/manual reconciliation without losing the payment.

[ ] Audit records include provider, transaction ID, state, and correlation ID.

[ ] No credential appears in source, documentation, logs, screenshots, or support tickets.

Provider

[ ] Test and production source IPs supplied.

[ ] Custom auth/signature/idempotency headers confirmed.

[ ] Clock is NTP synchronized.

[ ] Retry reuses the same transaction ID and idempotency key.

[ ] Provider has tested customer not found, invalid signature, timeout, duplicate, and payment status flows.

[ ] Provider technical and incident contacts supplied.

Activation gate

Production traffic starts only after both sides sign a test result containing:

- test customer code;
- provider transaction ID;
- SuperOptic correlation ID;
- payment amount;
- final payment state;
- UTC and Asia/Baku timestamps;
- explicit rollback owner.